CLOZAPINE

Agranulocytosis — mandatory blood monitoring

Myocarditis, constipation

CHEMICAL NAME 8-chloro-11-(4-methylpiperazin-1-yl)-5H-dibenzo[b,e][1,4]diazepine

CLASS Atypical antipsychotic (dibenzodiazepine) — reserved for treatment-resistant schizophrenia

TRADE NAMES Clozaril · Denzapine · Zaponex · Leponex

THERAPEUTIC USES / INDICATIONS

- Treatment-resistant schizophrenia (failure of ≥2 antipsychotics).
- Schizophrenia with severe tardive dyskinesia or EPS intolerance.
- Psychosis in Parkinson's disease (low dose).
- Reducing suicidal behaviour in schizophrenia.

ADMINISTRATION

- **PO**; registration with monitoring service and normal WBC/ANC before starting.
- Slow titration under supervision (BP, HR, temperature).
- If ≥48 h of missed doses, re-titrate from 12.5 mg.

DOSAGES

- **Start:** 12.5 mg once or twice on day 1, 25–50 mg day 2, then increase by 25–50 mg/day.
- **Target:** 200–450 mg/day (max 900 mg) in divided doses.
- **Parkinson's psychosis:** 6.25–50 mg at night.
- Target plasma level ≥350 ng/mL.

PHARMACOKINETICS

- **Peak:** 2.5 h.
- Metabolised by CYP1A2 (smoking ↓ levels by up to 50%), 3A4.
- **t½:** 12 h.

PHARMACODYNAMICS

- Weak **D2**, strong **D4**, **5-HT2A**, **α1**, **H1**, **muscarinic** antagonist (but M4 agonist → hypersalivation).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Hypersalivation (especially at night), sedation.
- Constipation (can be severe).
- Weight gain, tachycardia, postural hypotension, fever (early).

ADVERSE EFFECTS (SERIOUS)

- Agranulocytosis/neutropenia (~1%); weekly FBC for 18 weeks, then fortnightly to 1 year, then monthly.
- Myocarditis/cardiomyopathy (first 2 months) — tachycardia, chest pain, fever.
- Gastrointestinal hypomotility → ileus, bowel obstruction, death.
- Seizures (dose-related); metabolic syndrome; NMS; pneumonia.

CONTRAINDICATIONS & CAUTIONS

- History of agranulocytosis or bone-marrow disorder; inability to undergo blood monitoring.
- Uncontrolled epilepsy; paralytic ileus; severe cardiac disease, myocarditis.
- Use with drugs causing agranulocytosis (carbamazepine, co-trimoxazole); long-acting depot antipsychotics.

KEY INTERACTIONS

- **Smoking cessation:** ↑↑ levels → toxicity (reduce dose).
- **Fluvoxamine, ciprofloxacin, erythromycin:** ↑ levels.
- **Carbamazepine:** agranulocytosis risk; avoid.
- **Benzodiazepines (IV):** respiratory arrest.

NURSING MANAGEMENT

- Check FBC result before dispensing each supply (traffic-light system).
- Monitor bowel habit daily; prescribe laxatives prophylactically.
- Monitor temperature, pulse, BP during titration; troponin/CRP weekly first 4 weeks (myocarditis).
- Metabolic and ECG monitoring; seizure watch.
- Ask about smoking changes.

PATIENT EDUCATION

- Never miss blood tests; report sore throat, fever, flu-like illness immediately.
- Report constipation (no bowel movement for 2–3 days), abdominal pain.
- Report chest pain, palpitations, breathlessness.
- Tell your team if you stop or start smoking or miss doses for 2 days.

KEY POINTS

- Gold standard for treatment-resistant schizophrenia.
- Agranulocytosis — mandatory FBC monitoring.
- Myocarditis, severe constipation, seizures.
- Smoking changes levels; re-titrate after 48 h missed.